

Tutorial 5 (Velocity analysis)

(Analytical method is to be used by default; the descriptions of the problems in the book are refined)

- **Problem 6-5:**

A general fourbar linkage configuration and its notation are shown in **Figure P6-1**. The link lengths, coupler point location, and the values of θ_2 and ω_2 for the same fourbar linkages as used for position analysis in Chapter 4 are redefined in **Table P6-1**, which is basically, the same as **Table P4-1** in **Tutorial 3**.

- Copy Figure 6-1 and indicate the instant centers of $I_{1,3}$ and $I_{2,4}$ on it.

For Row a assigned, based on the solutions for **Problem 4-7** in **Tutorial 3**, and only for the open configuration of the linkage,

- calculate the angular velocities of Link 3 and Link 4;
- calculate the velocity of point P and sketch it in the figure, showing its magnitude and direction.

- **Problem 6-7:**

The general linkage configuration and terminology for an offset fourbar crank-slider linkage are shown in **Figure P6-2**. The link lengths and the values of θ_2 and ω_2 are defined in **Table P6-2** which is based on **Table P4-2** for **Problem P4-10** in **Tutorial 3**.

For Row a assigned, based on the solutions for **Problem 4-10** in **Tutorial 3**, and only for the open configuration of the linkage, calculate the velocities of the pin joints A and B and the velocity of slip at the sliding joint; sketch the velocities of pin joints A and B in the figure, showing their magnitudes and directions.